LITERATURE EVIDENCE REPORT — REPRODUCIBLE RAPID SCOPING REVIEW

Avionics and Payload Missions in Student Rocket Competitions

Near 3 km apogee: 3K and 10,000-ft student competition classes

Evidence to inform the RTC 2027 Student 3K Mission Design Report (MDR)

Search and verification cut-off: 5 November 2026

Abstract

Objective. Identify documented flight computers, sensors, recovery electronics, scientific payload missions and validation practices in student competition rockets within a measured or nominal 3-km apogee class, and map that evidence to RTC 2027 MDR claims. **Search.** The user's first broad Web of Science Topic strategy combined rocket/launch-vehicle/missile terms with avionics, navigation and telemetry terms; exports were labelled 1609 Web of Science and 788 IEEE Xplore records. **Methods.** The unique supplied files contained 2396 parsable BibTeX records (1609 WOS and 787 IEEE); 75 DOI/title duplicates were removed, leaving 2321 unique records. A documented broad title/abstract filter yielded 1148 candidates; 121 met a prioritisation score. Supplementary student-team technical reports, theses, CanSat mission guidance and design reviews were also assessed separately from the database denominator. **Results.** Seven competition/competition-linked examples and six contextual avionics sources were examined. AVES/AVES II anchor the 3-km avionics precedent, while a supplementary academic-first payload search documents scientific missions ranging from thermal/structural sensing and fluid dynamics to biomedical studies, imaging and CubeSat technology demonstration; no source establishes Pico superiority or verifies the proposed RTC thermal mission. Graphs show the screening profile, bibliographic topic signals, evidence classes and a primary reported hardware inventory; a separate battery curve is clearly labelled as an engineering model. **Limitations.** This is a rapid systematic *scoping* review, not a completed PRISMA systematic review: most candidates were not independently full-text screened, and nominal target altitude is not necessarily achieved apogee. **Conclusion.** The evidence supports a modular sensing, logging and redundant-recovery design *concept*, while actual Pico flight readiness must be established through simulation and test.

1 Review question and scope

Question: What avionics designs and science/payload mission concepts have been documented in student rocketry competitions with nominal apogees at or near 3 km, and what validation evidence exists for their use?

Apogee eligibility is explicitly classified: **A** = measured apogee $\leq$ 3000 m; **B** = nominal 3000 m competition class (actual flight apogee unspecified); **C** = a documented lower-altitude competition (actual individual flight apogee unknown); and **D** = relevant sounding-rocket research lacking sufficient altitude or competition evidence, retained only for contextual synthesis. In the 2026/2027 RTC "3K" category, the nominal goal is at least 3 km; that categorisation does not imply every entry's actual apogee was exactly or below 3000 m [1]. We include the 10,000-ft (3048-m) IREC class as a *near-3-km adjacent competition stratum*, not as a measured apogee $\leq$ 3000 m.

2 Methods

2.1 Review design, eligibility and research questions

This is a *rapid, systematic scoping evidence review*, not a completed two-reviewer systematic review or meta-analysis. It asks which flight computers, sensors, recovery-control arrangements, power systems, data logging and telemetry architectures were *documented* for student competition rockets whose nominal class or measured apogee is at or near 3 km. A further question is which directly documented design features can support RTC 2027 MDR requirements, rather than which commercial board is "best." The review uses PRISMA 2020 and PRISMA-S as *reporting guidance*, while disclosing incomplete screening and the absence of a prospectively registered protocol [2], [3].

Eligibility. Include as primary evidence: a named student competition/team, a measured flight altitude $\leq$ 3000 m, credible nominal 3-km class, documented lower-altitude event, or a 10,000-ft adjacent class, and at least one citable avionics design feature or outcome. Class A: measured flight $\leq$ 3000 m; B: nominal 3-km class with unconfirmed measured apogee; C: documented lower-altitude competition context; D: informative rocketry research without the required competition/altitude link (context only). Class E is an adjacent 10,000-ft (3048-m) IREC target category, which is only 1.6% above 3 km and is analysed separately, never relabelled as a $\leq$ 3000-m flight. Preflight technical reports are not proof of flight success. Exclude from primary cases orbital launchers, general military missiles, unrelated aircraft/UAV systems, and avionics papers with no competition/altitude evidence.

2.2 First search: databases, exact strategy and export counts

The **first supplied search** was deliberately sensitive and broad; the original Web of Science Topic expression is reproduced verbatim below (line breaks added only for typesetting):

```
TS=((rocket* OR "launch vehicle*" OR "sounding rocket*" OR missile*)
AND (avionics OR "flight computer*" OR "guidance system*"
OR "navigation system*" OR "flight control system*"
OR telemetry OR "attitude determination"))
```

This initial search yielded the user's stated **1609 Web of Science** and **788 IEEE Xplore** records. The field tag `TS=` belongs to Web of Science; the precise IEEE Xplore interface-specific field translation/limits were *not preserved* in the uploaded metadata, so no exact IEEE query is invented here. These counts are database-search *labels*, not assertions that every record pertains to competition rocketry. The Web of Science export carries 3 November 2026 retrieval metadata; the IEEE export is labelled as the November 2026 collection. No unrecorded date/language limits are presumed. A complete future review should also search Scopus, aerospace conference repositories, EuRoC and IREC team reports, university theses and official competition reports with a separate competition-specific strategy. Such additional retrieval has not been counted as part of the original database-search totals.

2.3 Bibliographic processing and transparent screening

Three uploaded files were inspected: `Combined 1609.bib`, `Combined IEEE 788.bib` and `Combined IEEE 788(1).bib`. The two IEEE files have identical SHA-256 hashes, so only one was processed. The Web of Science file contains 1609 parsable entry starts; the IEEE file contains **787 actual BibTeX entry starts** despite "788" in its filename (a further `@` is part of a conference title). The processing sample is thus $1609 + 787 =$ **2396** parsed records. Titles, abstracts, source labels, year and DOI were extracted. Deduplication preferentially used normalised DOI, otherwise a lower-case, punctuation-stripped title. This removed **75** records and retained **2321** unique bibliographic records.

The post-export *computational broad-screen proxy* required both a rocket-related expression and an electronics-related expression within combined title and abstract fields. The exact case-insensitive Python regular expressions are archived in the accompanying screening script; the term stems are also shown below.

> **Rocket set:** rocket, sounding, launch vehicle, IREC, EuRoC, TEKNOFEST, spaceport.
> **Electronics set:** avionics, flight computer, electronics, telemetry, data logging, parachute, recovery, apogee, flight control, sensor, attitude, control system, navigation, payload, onboard, embedded, altimeter, GPS, tracking.

Exactly **1148** matched; **1173** were non-matches. Among matches, an additive metadata heuristic (3 points for competition/student keywords in the title, 2 for avionics-related title terms, 2 for competition keywords in the abstract, 2 for altitude/apogee mentions in combined title/abstract; score threshold ≥ 5 of 9 possible points) identified **121 higher-priority candidates** (score ≥ 5). **This was not a comprehensive full-text screening of 121 studies**; the remaining 1027 broad candidates were not individually assessed in full.

Primary competition reports and institutional theses/webpages were additionally found through targeted searches and explicitly labelled as *supplementary sources*, not silently counted inside the 2321 indexed records. Seven competition or competition-linked cases were extracted for the descriptive evidence table; six further studies were retained as secondary contextual evidence. The seven are not the final included sample from a complete PRISMA screening, nor evidence of the population prevalence of any hardware model.

2.4 Focused supplementary search: payload missions and full-text access (5 November 2026)

A second, *targeted and separately tracked* public-source search was added because the original avionics keyword strategy was not designed to recover scientific payload studies. This expanded stage prioritised peer-reviewed academic papers and university repositories, next university capstone/technical reports and student-team primary pages, and lastly competition press reports. It searched for 3-km EuRoC, IREC 10,000-ft and RTC 3K cases, requiring identifiable mission objectives and separating flight-carried hardware from completed scientific outcomes. Queries combined `EuRoC`, `IREC`, `student rocket`, `CanSat`, `payload mission`, `temperature`, `pressure`, `imaging`, `experiment`, and `full text/PDF`. Searches were conducted in public search-engine results and checked against institutional repositories, university-team reports, open journals and competition document archives. This is a *purposeful supplemental evidence stream*, not an exhaustive new database search; its results were not added to the 2321-record denominator, and no invented PRISMA retrieval count is assigned to it.

Eligibility for *direct payload precedent* required a named rocket-team or competition technical report and a describable mission objective. *Adjacent payload precedent* includes CanSat guidance and team CDRs; these do not demonstrate

3-km rocket-flight performance. Peer-reviewed sounding-rocket electronics papers with no competition/altitude verification remain contextual. We classify access as **F** (full text actually read), **A** (abstract/metadata only), or **U** (full text not currently retrievable). A report's proposed device and mission are not equivalent to a verified flight outcome. Data extraction records mission goal, instruments, power/data interfaces, claimed mission outcome, altitude basis, and applicability to RTC Appendix 03, Section 7.1 [1].

Source checked	Access	What was read / remaining limitation
AVES 2021 [4]	F	Public team technical report; near-3-km mission, student CanSats and atmospheric measurement concept. Preflight objectives do not demonstrate completed science data.
AVES II 2022 [5]	F	Public report, Section 3.8; identifies three student 1U CubeSat projects. Detailed objectives for each project are not adequately described in the accessible main-report pages.
ILLUSTRIA thesis [6]	F	Full university thesis located at an alternative repository URL; PCB, state-machine and HIL sections accessible. Not evidence of a completed RTC-class flight.
Arduino Rocket Flight Computer [7]	F	Open six-page article with flight-data acquisition and postflight analysis; small model rocket, not 3K competition validation.
Netzel et al. 2026 [8]	F	Open journal article, including STM32F407 hardware, 32-byte records, and hardware-in-the-loop validation. No verified 3K competition flight.
UK CanSat guidelines [9]	F	Open competition document: 1-Hz air temperature and pressure, Pico/BMP280 example, mission planning and plots. Adjacent educational payload standard only.
PWr Aerospace CDR [10]	F	Public CanSat team design review: pressure, temperature, GPS, attitude and cameras. Planned hardware, not demonstrated RTC flight success.
Bakirci and Ozer 2022 (avionics) [11]	A	Abstract describes 10,000-ft scientific payload and redundant avionics; no accessible full paper found; no detailed part numbers attributed.
Bakirci and Ozer 2022 (recovery) [12]	A	University repository marks full text closed access; abstract supports only a general dual-stage recovery concept.

Access boundary. An inaccessible paper is retained with its access status rather than silently treated as full-text evidence. Institutional and team reports are primary *design disclosures*, not peer-reviewed performance estimates. Changes to full-text access can be recorded in a future update.

2.5 Data extraction, figure coding and appraisal

The extraction fields were source/venue, competition, nominal versus measured apogee, MCU, inertial/pressure sensing, telemetry/GNSS, persistent data storage, recovery path, available testing/outcome evidence and limitations. Reliability class **R1** denotes a primary technical report or thesis, **R2** an experimental/HIL research paper, **R3** an institutional story, and **R4** abstract-only metadata. Inconsistently reported or absent items were treated as *not disclosed*, not as absent hardware. No estimates of failure rates or processor "win rates" were made.

All graphs report their denominator and evidence basis. Workflow and year counts come from the two unique metadata exports. Figure 3 counts *non-mutually-exclusive regular-expression signals*, not confirmed use of components in flight, and may include military/large launch-vehicle papers because the first search explicitly included `missile*`. Figure 4 summarises only the seven manually extracted competition cases. Figure 5 uses an identifiable published component inventory in the *AVES II* technical report. A separate analytic power model (Fig. 7) is marked as *calculated engineering assumption*, not a flight measurement or an outcome extracted from the review [1], [5], [13].

2.6 Reproducibility and departures from a complete SLR

The accompanying package contains a screening script, the source-record audit and figure data tables (CSV). To independently reproduce the initial numbers, use the *original* two BibTeX exports: the ZIP does not redistribute those large, user-supplied libraries. The included seven-case charts describe the original core sample; additional targeted 10,000-ft examples below are NOT part of that seven-case denominator. The current narrative/graphs represent an honest rapid evidence review; a journal-grade systematic review would require independently screened full texts for all potentially eligible records, exclusion reasons, search-platform settings, a registered protocol or predeclared amendments, and a completed PRISMA flow [2], [3].

2.7 Search and screening profile (PGFPlots)

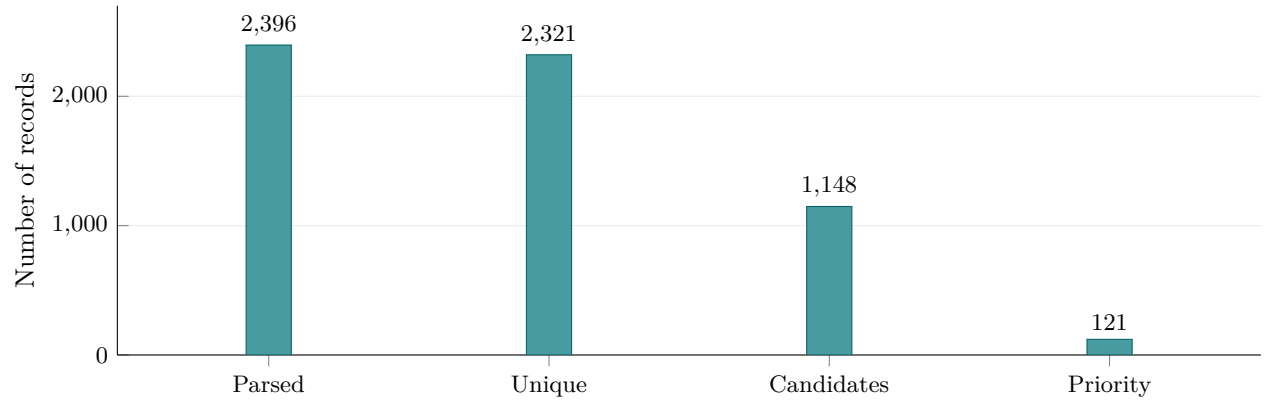


Figure 1: Sequential *metadata-screening stages*: parsed records, DOI/title-deduplicated records, broad keyword candidates, and heuristic-priority candidates. The last bar is *not* a full-text inclusion count. Data: exported bibliography and reproducible script.

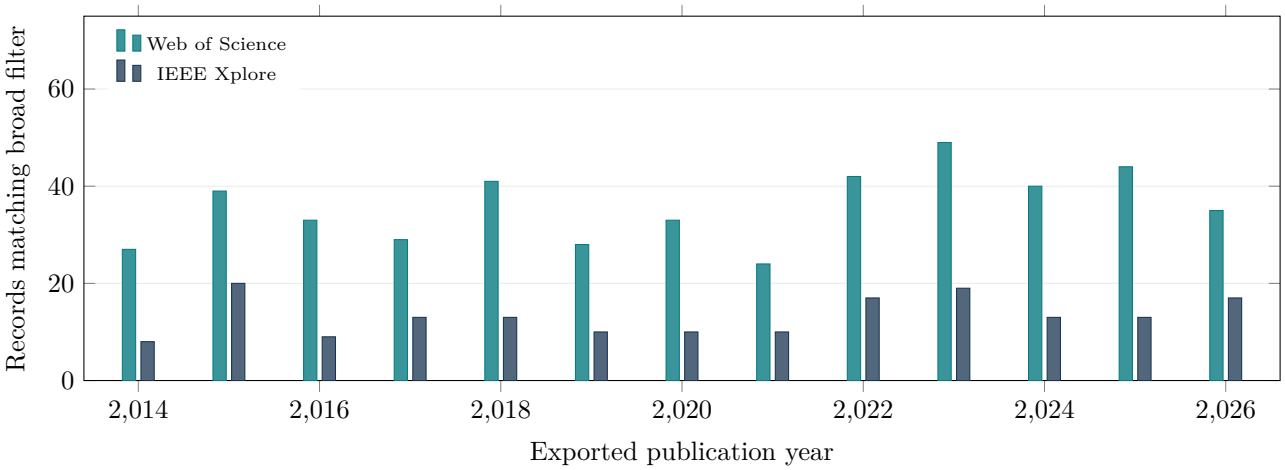


Figure 2: Publication-year distribution for the *636 of 1148 keyword-filtered records dated 2014–2026*, by retained source; records outside this window are not plotted. This shows the broad initial literature landscape and *does not* estimate the number of qualifying 3-km student flights. Data: `figure_annual_counts.csv`.

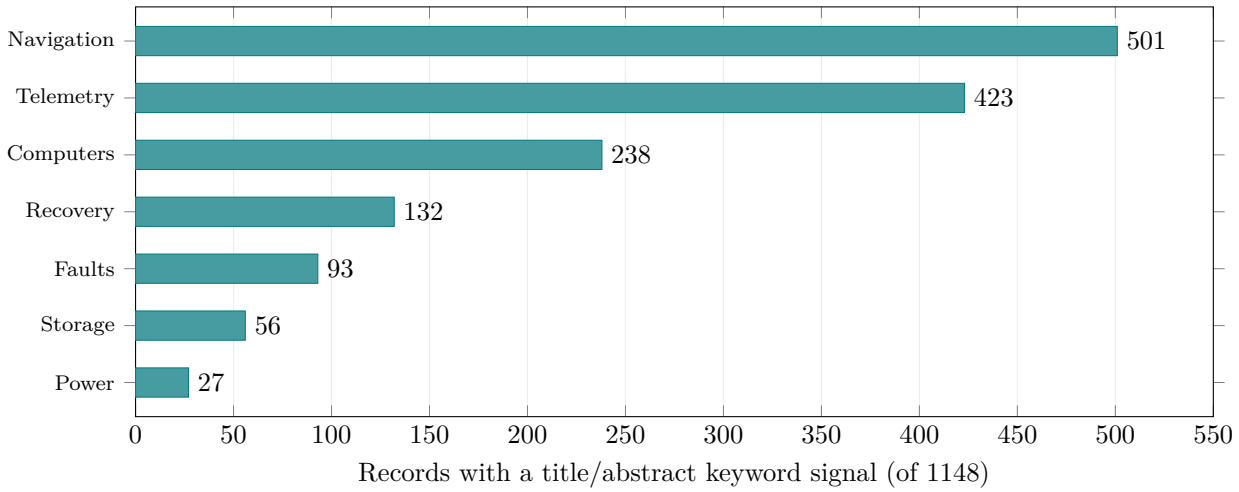


Figure 3: *Non-exclusive lexical signals* in the 1148 broad candidates: navigation/IMU/barometric/GPS (501), telemetry (423), flight computer/avionics (238), recovery/deployment (132), fault tolerance/redundancy (93), storage/logging (56), power/energy (27). Counts use documented regular expressions, not actual installed-hardware frequencies.

3 Results: competition cases inside or nominally at 3 km

Team / source	Apogee eligibility	Documented avionics hardware and functions	Evidence / limitations
ASTG *AVES* (EuRoC 2021) [4]	B: 3000-m target class	Custom dual-core ARM flight computer; 6-DOF IMU, separate high-g accelerometer, barometer, two GNSS receivers, SD and flash memory, RavenOS, telemetry; COTS flight-computer path and two-stage recovery.	R1 team technical report; 3000 m is the mission target, not independently verified recorded apogee.
ASTG *AVES II* (EuRoC 2022) [5]	B: 3000-m target class	STM32H745ZI (Cortex-M7/M4), custom flight computer and RTOS RavenOS, telemetry, sensor redundancy, black box/cameras; SRAD and COTS recovery-control paths; Eggtimer Quantum used for official logging rather than redundant recovery.	R1 detailed pre-flight technical report including ground-test annexes; team reports a completed EuRoC flight, but numeric flight apogee is not established here.
CTU *ILLUSTRIA* flight-computer study (2024) [6]	B: linked to nominal 3-km EuRoC entry	Custom STM32H743 board; pressure/inertial sensing, CAN/SPI/I^2C, PX4/NuttX state machine, MAVLink and software-/hardware-in-loop methodology.	R1 university thesis; bench/HIL design validation, not proof the same board completed a 3-km flight.
Zapadlo's 2023 Czech Rocket Challenge computer [14], [15]	C: lower-altitude event (organiser presently lists 400–800 m typical)	STM32 flight computer with LoRa; battery/supercapacitor selector, redundant-power design and telemetry.	R4 repository abstract; verified on stratospheric balloon, not asserted as rocket flight-proven. 2026 category figures used as approximate context only.
Kadlec and Walla model-rocket project (Czech Rocket Challenge experience) [15], [16]	C: lower-altitude competition context	Barometric altitude, acceleration, attitude, temperature and GNSS; onboard logging, image capture, real-time ground communication and autonomous operation after link loss.	R1 published student proceedings; specific project apogee and detailed microcontroller not established from reviewed portions.
Tamkang *Stargazer* demonstration at RTC 2026 [17]	A: reported 2956 m measured	New-generation avionics were tested on the demonstration flight; organisers' motor used.	R3 institutional news report; genuine altitude/outcome, but no published sensor/MCU list in source. Not presented as a competing finalist.
National Formosa University *Skybound Singularity*, RTC 2026 [18]	B: RTC Student 3K category	Recovery and payload deployment featured in the mission; electronic system detail and individual MCU/sensor models not substantiated by this source.	R3 university results report; 3K category and award confirmed; exact apogee and full avionics bill unavailable.

3.1 Evidence coverage and a directly documented hardware example

3.2 Interpreting the primary competition evidence

Convergent functions: (i) pressure and inertial sensing for flight phase recognition; (ii) logging independent of ground communications; (iii) parachute deployment control separated from at least some non-safety subsystems; (iv) telemetry or GPS for recovery and post-flight analysis; (v) explicit ground or simulation verification. The best documented 3-km team reports are AVES and AVES II; the expanded payload cases are a distinct purposive sample and not evidence that their exact parts are universally optimal [4], [5].

Architectural diversity: 3-km AVES II uses a custom high-performance STM32 MCU plus separate COTS electronics; the Czech challenge thesis documents an STM32-plus-supercapacitor arrangement; the RTC Stargazer report demonstrates an actual near-3-km flight but does not expose its part numbers. Consequently, there is no defensible numerical frequency table of microcontroller brands for all 3-km competition finalists [5], [14], [17].

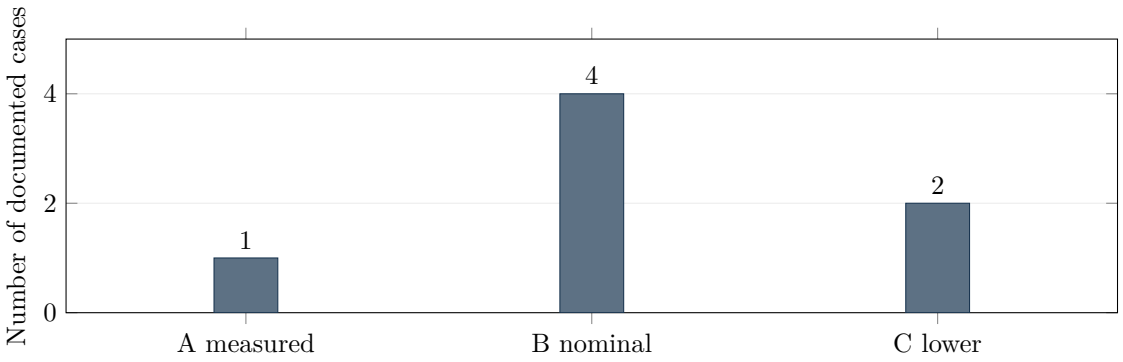


Figure 4: Eligibility status of the *seven manually extracted competition/competition-linked examples*. Only one has a published measured apogee within 3 km (Tamkang's demonstration); four are nominal 3-km-class projects, and two are lower-altitude competition-linked projects. No prevalence, reliability or success-rate inference is warranted [4], [5], [6], [14], [16], [17], [18].

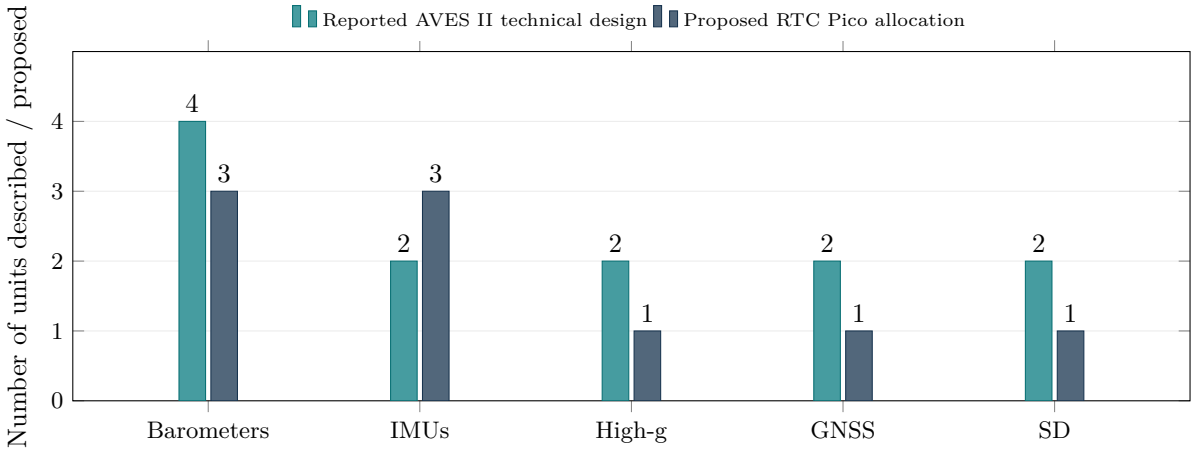


Figure 5: **Descriptive design comparison, not performance evidence.** AVES II's published preflight inventory reports 2 BMP390 plus 2 MS5607 barometers (4 total), 2 BMI088 IMUs, 2 H3LIS331DL high-g sensors, 2 SAM-M8Q GNSS receivers, and 2 SD cards [5]. The proposed RTC allocation includes 3 barometers, 3 MPU6050 IMUs, 1 baseline ADXL375 high-g sensor, 1 organiser GPS beacon and 1 logger SD; this proposal is *unqualified*, with separate FRAM checkpoints excluded from the SD count. Counts describe architecture only; higher counts do not establish greater reliability.

4 Contextual avionics research: altitude or competition unverified

These studies appeared in, or are strongly linked to, the uploaded literature libraries. They *support engineering reasoning* but were *not included as verified ≤ 3-km competition flight cases.*

Study	Relevant evidence	Why secondary
Netzel et al. (2026), *HardwareX* [8]	STM32F407, high-g ADXL375, IMU, MS5607 barometer and non-volatile logging, with end-to-end hardware-in-the-loop (HIL) phase/event testing.	No demonstrated competition flight; published HIL study.
Pathak et al. (2022), S3FC [19]	Launch interlocks, filtering against false deployment, GPS downlink and CRC for telemetry integrity.	Exact apogee/category absent in available abstract.
Kahe (2017) [20]	COTS-based dual-redundant sounding-rocket flight computer.	No verified 3-km competition altitude.
Moschidis and Bithas (2022) [7]	Model rocket using accessible Arduino flight-computer components and simulation-to-flight comparison.	Specific student competition and target apogee unverified.
Bakirci and Ozer (2022) [11]	Scientific experiment unit integrated with subsonic rocket avionics, telemetry and recovery functions.	Competition or altitude unverified.
Andrade et al. (2022) [21]	Heltec LoRa 32 proof-of-concept sounding-rocket telemetry.	Proof of concept; no verified competition flight at ≤ 3 km.

5 Expanded review: what payload missions do other near-3-km student rockets undertake?

Evidence priority and scope. First preference is given to scholarly conference publications, university repositories, capstone reports and institutional projects. Where these do not specify a concrete mission, the review uses traceable university-affiliated team documentation and then competition/news sources. These additional records were found by a *purposeful supplementary search*, not a comprehensive independent second screening of all 2321 unique database records. A carrier in the 3000-m class may overshoot 3000 m. The IREC 10,000-ft class (3048 m) is treated as an adjacent comparator, not as a rocket that necessarily reached exactly 3 km. *Flight performed*, *hardware carried*, *post-flight data recovered*, and *scientific hypothesis validated* are **different evidentiary claims** [1], [22], [23].

5.1 Academic publications and university repository evidence (highest priority)

Academic source / class	Payload type	Mission objective and measurement principle	Evidence / limitation
Seoul National University, *HARANG*, IREC 2023, 10,000 ft [23]	CubeSat deployment / technology demonstration	Deliver and deploy a 3U CubeSat using a student-developed sounding rocket. The conference abstract discusses onboard telemetry and a separation interface; it does *not* state a separate CubeSat science experiment.	Peer-reviewed conference abstract; launched, premature nosecone opening suspected; experiment outcome unreported.
University of Maryland, *Terpulence II*, IREC 2022, 10,000 ft [24]	Capillary fluid management	Test a liquid-tank geometry in which capillary action moves a water analogue, with the goal of understanding propellant-management methods without ullage thrusters.	Institutional technical report; mission described; post-flight science result not established from repository summary.

Source / class	Type	Mission and measurement (continued)	Evidence
University of Georgia, *Cloud / I.M.P.U.L.S.E.*, IREC 2025, 10,000 ft [25], [26]	Liquid slosh / direct surface sensing	Observe fluid free-surface motion in a contained tank using a downward-facing LiDAR, Raspberry Pi-based acquisition and comparison with a CFD model.	University capstone/competition reports describe flight integration; publicly reported objective, not independently reproduced raw dataset.
University of Georgia, *Magnolia / HEART*, IREC 2026, 10,000 ft [26], [27]	Biomedical fluid dynamics	Test a mechanical heart valve in a closed-loop simulated-blood circuit under rocket acceleration, with flow and temperature sensing. A Pico W ran control and measurements at 0.05-s intervals.	University capstone account reports preliminary leakage tests and flight data acquisition; medical effectiveness is *not* thereby demonstrated.

Supporting scholarly payload-engineering research, outside the competition-flight inclusion group. Moura and colleagues' peer-reviewed MicroGravity Explorer Kit work proposes modular processing and experiment modules with configurable sensors, imaging and actuators; it is a useful *design methodology*, not a documented RTC-class flight or a validation of the team's Pico payload [28]. Duraisamy's Embry–Riddle doctoral thesis reports a systematic payload engineering approach involving requirements, CAD/FEA, vibration and load testing, and CFD for liquid payloads, but is not proof of a particular 3-km competition mission [29]. These sources strengthen the requirements-to-experiment methodology rather than inflating the number of qualifying flown payloads.

5.2 University-affiliated EuRoC projects and flight reports

Team / flight class	Payload class	Reported objective / hardware	Flight / outcome status
ASTG *AVES*, EuRoC 2021, S3 [4]	Atmospheric and student CanSats	Up to three CanSat-form-factor experiments, some prepared by collaborating technical schools for environmental observations.	Target 3000 m; mission design documented; science outputs not established here.
ASTG *AVES II*, EuRoC 2022, S3 [5]	Educational 1U CubeSats	Three 1U payload projects (Hawfinch, Buffalo–REVD, Accipiter) in a hosted student payload module.	Named projects documented; detailed measured outcomes not independently established.
TU Munich/WARR *EX-4B / D.I.N.O.labs V2*, EuRoC 2024, L3 [30]	Photobiology / bioluminescence	Examine bioluminescent algae response to altered acceleration and shear, with eight live samples and light-sensing apparatus.	EuRoC Payload Award; team reports payload recovered functional, but quantitative biological findings not supplied.
TU Darmstadt/TUDSaT *FRoDO-M*, EuRoC 2024, S3 [31]	CanSat video + structural measurement	Four carried CanSats, including a 2U live-video payload; rocket's instrumented "smart fins" use strain gauges to study aerodynamic deformation (vehicle instrumentation rather than necessarily a separate payload).	Team reports achieved 2993.7 m; carried payloads. Individual CanSat datasets not disclosed.
KSat Stuttgart / rockit Karlsruhe *FROG*, EuRoC 2025, 3-km class [32]	Ferrofluid physics / imaging	Camera-based observation of magnetically shaped ferrofluid and a magnetic-spring deflection; Raspberry Pi Zero, Pi camera, accelerometer and dedicated battery management.	Public design description; completed flight and usable results not verified in cited source.

5.3 Additional public rocketry team reports (secondary evidence)

Rocket / competition	Payload function	Reported objective	Strength / limit
RED *Baltasar*, EuRoC 2022, S3 [33]	Hosted CanSat/PocketSat experiments	Carry one 1U CanSat and one 4U PocketSat in dedicated compartments for educational science and technology demonstration.	Team reports 3006-m apogee; exact experimental data not disclosed.
RED *Camões*, EuRoC 2023, S3 [34]	Biological science / spectroscopy	Three retained experiments involving bacterial reflectance spectroscopy, tardigrade viability and stem-cell survival.	Team reports 3014.89-m apogee and payload award; no validated biological results provided.
STAR UC3M *Lince*, EuRoC 2023, 3-km class [35]	GNSS/SDR and flight technology	Test a high-dynamics GNSS software-defined receiver accelerated by FPGA and an experimental aerobrake system; three in-house payloads reported, not all specified.	Team reports 3631 m and recovery of deployable payload; data quality not quantified.
STAR UC3M *Pícaro*, EuRoC 2024, 3-km class [36]	Return-to-launch CubeSat technology	Demonstrate a CubeSat-format payload with a Return-To-Launch (RTL) concept.	Team reports 3183 m; payload lost, so successful payload mission not established.
UCF *Odysseus / Sunflower*, IREC 2026, 10,000-ft class [37]	Autonomous solar pointing	Carry a 3U self-orienting solar-panel system ("Sunflower") integrated in the nose cone.	Team reports 9890-ft flight; self-orientation experiment success not demonstrated by cited page.
RTC 2026 Student 3K concepts (unnamed teams) [38]	Video, autonomous boat/drone	Reported proposals included real-time imagery, a nosecone-deployed boat intended to return after splashdown, and a drone deployment concept.	Competition reporting confirms proposals, not which team completed each or their test results.

Interpretation. These cases span *environmental measurement, structural health monitoring, imaging, GNSS technology verification, fluid mechanics, life-science experiments, autonomous energy harvesting, deployable recovery and CubeSat hosting*. They should not be pooled as equally strong "successful payload missions." For example, a team reporting a flown rocket does not show that the experiment returned analyzable scientific data. Nor should a rocket's mandatory pressure sensor, airbrakes or recovery controls automatically be counted as the *separate additional mission payload* required by RTC [1].

5.4 Measured apogees of selected university-team examples

Interpretation of Figure 6: these are the *authors'/teams' reported achieved rocket apogees*; they are *not* independent telemetry validation, payload operating altitude or payload experiment success. They illustrate why nominal 3-km category rockets may achieve slightly below or above 3000 m.

5.5 Scientific design patterns and feasibility for RTC Student 3K

Table 6 translates the external missions into a design trade-off. *Complexity* is an explicit qualitative **team-engineering judgment**, not a published outcome or comparative price estimate. Safety, power, mechanical retention and the revised RTC Appendix 03 Section 7.1 constraints must be checked separately: EuRoC/IREC may permit deployment or stored-energy systems that are not necessarily allowed for RTC.

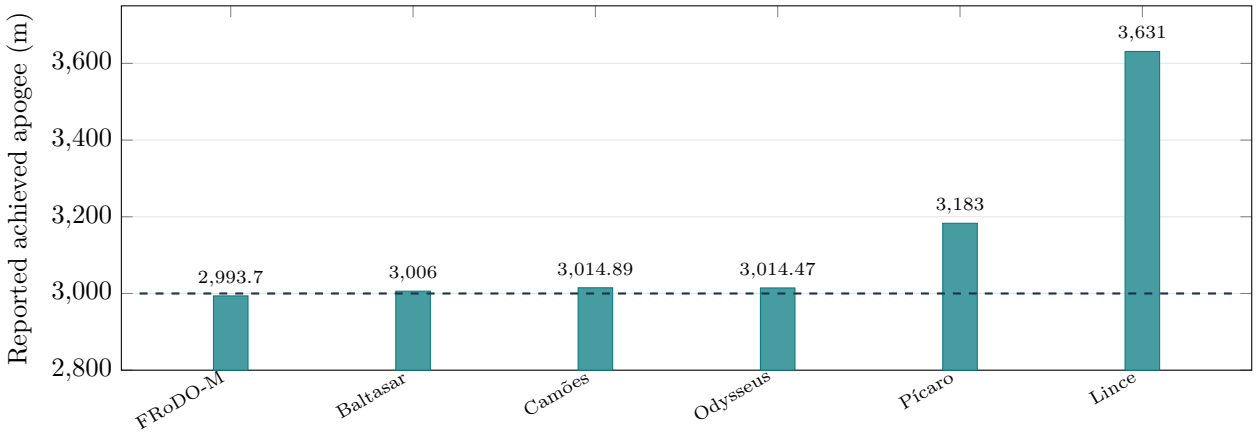


Figure 6: **Selected self-reported flight altitudes, not mission-payload performance.** FRoDO-M [31]; Baltasar and Camões [33], [34]; Odysseus (9890 ft $\times 0.3048 = 3014.47$ m) [37]; Pícaro and Lince [35], [36]. Cases were purposively selected and are not a census. Dashed line is the nominal 3000-m target.

Table 6: Evidence-informed mission options for RTC; feasibility judgments are provisional.

Mission category	Example from documented teams	Potential verifiable output	RTC implications / effort
Two-zone thermal/ambient environment	AVES/CanSat measurements; FRoDO-M instruments [4], [31]	Temperature difference vs flight phase, paired-sensor calibration, uncertainty and complete time series.	Low complexity; novelty relies on a testable scientific question; retained and no radio possible.
Structural response and vibration	FRoDO-M strain-gauge "smart fins" [31]	Strain or acceleration spectrum versus vehicle flight phase, compared with structural model or ground baseline.	Low–moderate electronics; installation, sensor calibration and structural interfaces matter.
Onboard imaging / live imagery	FRoDO-M video CanSat, RTC video proposals [31], [38]	Time-stamped imagery plus image completeness, field of view and storage/telemetry evidence.	Moderate complexity; SD bandwidth, power and splashdown protection; live RF requires approval.
Fluid sloshing / capillarity	UGA I.M.P.U.L.S.E. and UMD CubeSat [24], [25]	Direct fluid-level/displacement history synchronized to acceleration and compared with theory/CFD.	Moderate–high; liquids, sealing, centre-of-mass effects and tests.
Biological / biomedical response	WARR algae, Camões biology and UGA HEART [26], [30], [34]	Optical activity or flow/temperature response relative to controls with handling records.	High; contamination, containment, temperature control, sample stability and authorisation.
Ferrofluid / magnet response	KSat FROG [32]	Visible peak geometry/magnetic-spring deflection versus measured acceleration.	Moderate–high; sealed fluid/glass, magnets, imaging and light control.
GNSS/SDR or solar pointing	Lince FPGA GNSS receiver; Sunflower solar-panel demonstrator [35], [37]	Receiver fix/tracking quality or measured panel angle and voltage/current versus reference.	High; more hardware, integration and calibration, but clear technology demonstration.
Deployment / return	HARANG CubeSat, UC3M Pícaro, RTC 2026 concepts [23], [36], [38]	Deployment and return data.	Very high; flight-safety and regulatory restrictions; approval needed.

A critical microgravity caveat. A 3-km rocket coasting ballistically does *not* automatically provide the

long-duration, low-disturbance microgravity environment obtained by dedicated high-altitude sounding rockets. Drag, spin and vibration alter specific force. A project labelled "microgravity" should report its measured acceleration/rotation environment before claiming a controlled microgravity exposure [24], [28].

5.6 Implications for the current proposed thermal payload

The documented peer designs offer stronger experimental comparisons than simply logging two temperatures. For the RTC Pico concept, a feasible **baseline** remains two calibrated, retained DS18B20 probes (avionics bay versus inner airframe wall) with phase-aligned data and uncertainty estimates. The new review suggests two optional, still team-proposed refinements: (i) compare the temperature difference to an independently measured acceleration/pressure history, and (ii) report a measured thermal response lag to a controlled preflight step input. These additions preserve relatively low mass/power and make a testable *engineering-science objective*, rather than a second instance of standard rocket telemetry [1], [9].

A more distinctive yet still plausible alternative is **structural strain or vibration measurement**, motivated by TUDSaT's smart fins, if the structures team can safely mount and calibrate gauges without affecting flight integrity. The fluid, biological and deployable missions have real university precedent but need substantially more verification and cannot be treated as inexpensive, RTC-compliant substitutions without explicit systems/safety review.

Recommended payload MDR evidence package: a one-sentence scientific question, a null/comparison condition, a measurement diagram, sensor range/accuracy, expected operating timeline, mass and power budget, retention/containment plan, measurable full/partial/failure criteria, planned ground verification and a specified method for recovering data after splashdown. The organiser's GPS beacon is distinct from the additional payload [1], [39].

5.7 Limits of the expanded payload search

The evidence set includes one peer-reviewed IAC abstract and a peer-reviewed platform-methodology paper, one university repository technical report, university capstone/team narratives, and secondary competition reporting. A few teams report recovered hardware but not raw data or analysed experimental outcomes. No pooled payload success rate or quantitative "best mission" ranking is warranted. The new cases are reported separately from the original seven-case avionics figures and the 2321 database-record denominator; their discovery does not imply that all possible near-3-km payload studies were found. The case studies provide *design possibilities and practical failure lessons*, not flight qualification of the proposed RTC equipment.

6 Evidence synthesis by avionics subsystem

Subsystem	Finding from case studies	RTC 2027 implication
Flight computer	Primary 3-km reports favour purpose-built MCU-based computers; STM32 explicitly documented in AVES II and ILLUSTRIA. Other lower-altitude designs use STM32/LoRa; selected small-model research uses Arduino. [5], [6], [7], [14]	Pico is a *candidate*, not literature-proven best choice; compare compute/I/O, power, failure modes, software support and qualification tests.
Altitude / motion	Inertial + barometric sensors occur repeatedly; AVES documents a separate high-g accelerometer, an important precedent for full boost-range logging. [4], [8]	Validate MPU6050 $\pm 16g$ against actual thrust/mass/shock envelope; specify barometer venting and sensor error handling.
Data logging	Local SD/flash or black-box recording decouples mission evidence from an intermittent radio link. [4], [5], [8]	Propose timestamped non-volatile flight record; demonstrate buffer occupancy and recoverable data after resets.

Subsystem	Finding	MDR implication
Recovery / redundancy	AVES II includes separate custom and commercial flight-computer paths; lower-altitude work examines alternate power sources. [5], [14]	Redundant Picos are an unvalidated *team design*; scrutinise common-mode firmware, barometer/pressure port and shared release mechanism failures.
Telemetry and position	Team implementations range from custom ground stations to LoRa, while RTC requires its standard GPS beacon. [1], [4], [14], [21]	Mandatory official GPS cannot be replaced with an unapproved radio; extra telemetry requires frequency compliance.
Testing	EuRoC technical reports include subsystem ground-test evidence; ILLUSTRIA and Netzel demonstrate simulation/HIL methods. [5], [6], [8]	Include pressure/flight-profile replay, safe recovery test loads, battery endurance, vibration, data integrity and integration testing in MDR verification plan.

7 Implications for the Raspberry Pi Pico MDR concept

The present team proposes one Raspberry Pi Pico as a non-safety logger/payload computer and two additional, independently powered controllers as potential recovery channels. This architecture is not claimed to have appeared in a verified 3-km competition report. Its justification should instead be written as a *traceable engineering proposal*: low MCU cost and I/O capability [13]; precedent for separate mission and recovery duties [5]; and documented need for deterministic event handling and HIL testing [8]. The final recovery arrangement must meet the specific 2027 RTC release and arming rules [1].

Decision	Evidence-supported rationale	Evidence still needed
Pico for logger	Low-cost, open ecosystem; separate acquisition and storage tasks can be implemented and benchmarked [13].	Measured sample jitter, storage stalls and energy use with production hardware.
IMU + barometer	Similar measurement functions appear in published competition systems [4], [5].	Peak-g simulation, pressure-port bias and calibration error.
Recovery A/B	Comparable reports use distinct recovery-computer paths [5].	Independent release-device feasibility, common-cause FMEA and repeated safe-load tests.
Local logs + GPS	Black-box data preservation is a documented approach [4], [8].	Retrieved official beacon data, timestamp alignment and storage integrity.
Payload acquisition	AVES school CanSats and adjacent CanSat competitions support a separate sensing mission, defined outputs and an integration plan [4], [9], [10].	Declared scientific objective, calibration and measurable flight success criteria.

7.1 Quantitative engineering check: proposed logger power and storage (model, not flight evidence)

The reviewed competition studies support the *architecture and functions*, not a numerical Pico endurance guarantee. To make the RTC electrical-endurance claim testable, a separate, explicitly *assumption-based* preliminary model uses a nominal 3.7-V, 2.6-Ah cell (9.62 Wh), 85% converter efficiency, and 3.3-V subsystem operating voltage. With a constant average current I (amperes), predicted runtime is

$$t_{\text{ideal}}(I) = \frac{3.7\,(2.6)\,(0.85)}{3.3\,I}\ \text{hours}.$$

The team design proposal assumes an average logger load near 56 mA but conservatively sizes around 200 mA; the predicted duration at 200 mA is about 12.39 h. This *is not a measured runtime*: battery capacity derating, cut-off

voltage, peak SD-card write current, protection losses, temperature, ageing and release loads must be tested. The RTC's > 1.5 h requirement applies to the actual assembled system [1], [13].

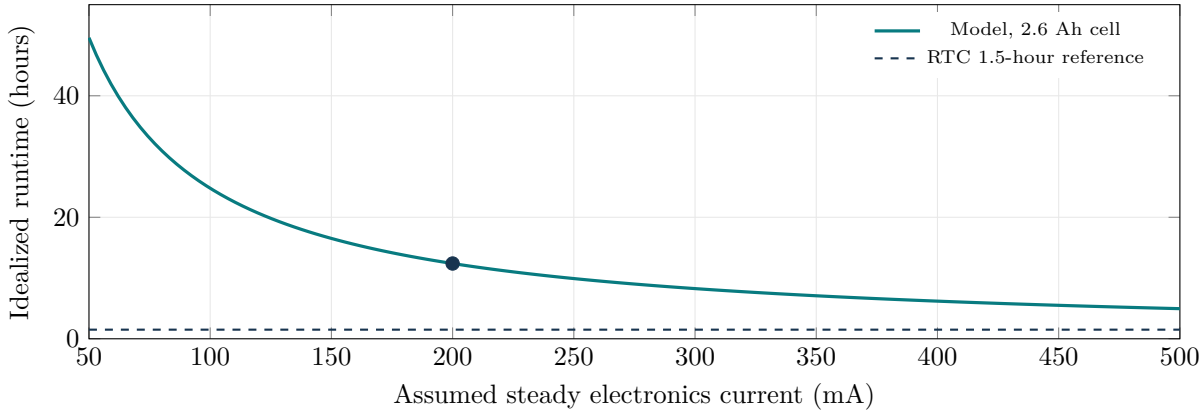


Figure 7: **Calculated scenario, not test results**. Predicted battery-endurance envelope for the proposed logger under stated assumptions. At 200 mA, predicted runtime is about 12.4 h; qualification still requires an actual > 1.5 h endurance test [1], [13].

Storage sizing (calculated). The revised MDR proposes a 44-byte fixed record at 200 samples/s, including IMU, high-g and two temperature channels; this gives $44 \times 200 = 8800$ B/s; across 90 min, 47.52 MB (decimal). This supports a preliminary memory-capacity choice, *not* evidence that microSD write latency, vibration retention or file-system reliability meets flight requirements. The latter must be measured with a worst-case logging test.

8 Adjacent 10,000-ft IREC competition stratum (3048 m)

The 10,000-ft IREC category is an especially relevant comparator to RTC 3K because $10,000\,\text{ft} \times 0.3048 = 3048\,\text{m}$, only 48 m (1.6%) above a nominal 3,000 m target. To avoid changing the original seven-case descriptive figures without rerunning the complete search, this additional stratum is presented *separately*, not added to Figures 4 or 5. The first broad indexed search lacked competition terms and was supplemented by team-project web reports, so neither figure should be treated as a comprehensive sample.

River City Rocketry "Sovereignty" (2026 IREC 10k COTS): the project description documents two commercial recovery electronics units (Blue Raven and StratologgerCF), independent GPS trackers (Featherweight and Telemetrum), locally recorded sensor data and a separate functional payload with its own power [40]. The website reports a *target* of 10,000 ft and a *predicted* 10,618 ft, so it is a design comparator, not evidence of successful flight or a measured achieved apogee. It supports discussion of functional and power-domain separation, but the pyrotechnic architecture described by this US competition entry cannot be assumed compatible with RTC 2027.

Implication for Pico MDR. AVES/AVES II directly motivate separate high-g measurement and redundant recovery architectures; 10,000-ft IREC projects provide an additional comparison for independent commercial recovery electronics and GPS paths. None demonstrates that three Pico development boards have flight-qualified reliability; the current proposal still needs simulation and bench/qualification tests [4], [5], [40].

9 Suggested MDR requirement–evidence map

RTC topic	Relevant published precedent	Planned evidence for MDR/PDR
Required flight data	AVES primary multi-sensor, logging and GNSS architecture [4]; RTC specification [1].	Data-flow diagram; rate/range table; bench data replay.
Drogue and main	AVES/AVES II two-stage recovery and redundant avionics [4], [5].	Proposed logic, two-path release diagram, fault analysis.
Power endurance	Alternate power-path design in lower-altitude competition thesis [14]; RTC electrical rule [1].	Per-domain power budget, battery protections and endurance plan.
Mission payload	AVES student CanSats and ILLUSTRIA sensor integration [4], [6].	Payload mission and measurable criteria; thermal calibration plan.
Verification	AVES II test appendices, ILLUSTRIA HIL thesis and Netzel et al. HIL [5], [6], [8].	Planned hardware-in-loop, safe-load, vibration and data checks.

10 Evidence-to-MDR claim audit: what the figures do and do not establish

MDR proposal	Directly documented supporting evidence	Proposed claim and unverified gap
Modular MCU flight computer	AVES II reports custom STM32H745ZI and distinct recovery/electronics functions; thesis documents STM32H743, software/HIL validation [5], [6].	Supports feasibility of the *functional architecture*; no published verification of three Picos for an RTC 3K flight.
Barometer + IMU + high-g sensing	AVES II reports duplicate barometers/IMUs and high-g accelerometers (Fig. 5) [5].	Validate BMP280 pressure ports and whether MPU6050 $\pm 16g$ clips during boost/deployment and whether a baseline $\pm 200g$ ADXL375 is adequate; model is not a sensor qualification.
Independent recovery paths	AVES II documents SRAD/COTS functions and redundant sensor paths [5]; separate technical source discusses redundancy [20].	Do not infer a recovery failure probability from team reports. Common pressure environments, firmware and mechanical release remain possible single-point failures.
Persistent flight data and GPS	AVES/AVES II describe flight-data storage, GNSS and telemetry [4], [5].	RTC official GPS beacon compatibility and access to the trajectory file remain team integration work.
Power $>$ 1.5 h and storage	Power curve (Fig. 7) and 47.52-MB revised storage need are conditional calculations; manufacturer Pico specification [13].	Not proof of real cell capacity, SD-card reliability, connector survival or launch loads. Full-power endurance and fault tests remain mandatory.
Payload experiment	AVES/AVES II hosted educational payloads; comparable university projects include WARR biology, UGA fluid and biomedical sensing, TUDSaT video and strain gauges, and published CanSat environmental missions [4], [5], [9], [25], [26], [30], [31].	The team's two-probe thermal mission still needs its own calibration, uncertainty analysis and measurable flight success criteria.

11 Evidence quality, limitations and conclusions

Evidence quality. Two team reports provide unusually rich 3-km detail but were prepared by the same organisation in successive years. One university thesis provides a 3-km design connection but not a verified flight of its new computer. Lower-altitude case studies are useful for power, telemetry and design development but operate in different loads and mission envelopes. A university news report gives actual near-3-km apogee with almost no reproducible avionics BOM. These weaknesses preclude a pooled success rate, failure probability or ranking of processors.

Review limitations. (1) A single reviewer applied the screening workflow; (2) 1027 broad candidate records were not all individually fully screened; (3) no protocol was prospectively registered; (4) publication and outcome reporting are highly variable; (5) multiple primary technical reports were published pre-flight, while flight outcomes appeared separately; (6) published web documentation can change; (7) nominal 3-km target is not the same as measured apogee ≤ 3000 m. To upgrade this work to a formal systematic review, independently screen all 2321 records (or a pre-registered refined search with documented exclusions), retrieve all eligible full texts, and provide a complete PRISMA flow and critical appraisal [2], [3].

Conclusion. Public evidence for approximately 3-km student competition avionics favours modular MCU-based design, calibrated barometric/inertial sensing, flight logging independent of telemetry, well-defined recovery redundancy and extensive ground verification. The expanded payload evidence also shows that student teams have undertaken environmental sensing, live imaging, strain measurement, liquid sloshing and capillarity, biological/biomedical experiments, GNSS/SDR demonstrations and autonomous technology demonstrations; the reported outcomes vary greatly in evidentiary strength. The evidence does *not* establish any particular board or payload mission as universally optimal. A Raspberry Pi Pico architecture can be defended in the RTC MDR as an economical preliminary design, provided the team specifies and eventually demonstrates its timing, sensor range, fault isolation, recovery authority, power margin and flight-data integrity against RTC requirements.

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